Milestone 2: Dataset Preparation

Socratic AI Tutor

Group 2 | Data Science and AI Lab, IIT Madras BS

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What we prepared

Dataset	Used for	Size after preparation
ASSISTments 2009-2010	Learner model (DKT)	256,060 answers, 3,408 students, 82 skills
MathDial + SocraTeach	Socratic model fine-tuning	37,413 conversations
NCERT Maths + Science	Course pack (study text)	3,164 text chunks, 10 books

ASSISTments and chat data prepared by Hariom Pandey; NCERT, packs, scenarios and validation splits by Sandeep Nalla.

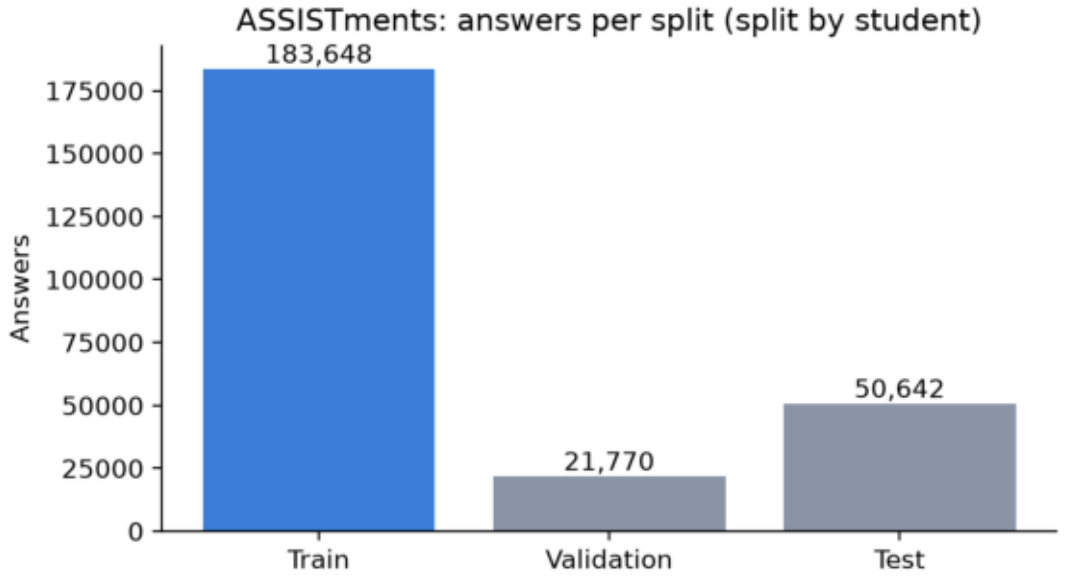
ASSISTments: learner model data

- 303,260 answers; kept 4 columns; removed 1 missing row, 46,668 duplicates, and students with under 3 answers

- 82 skills; each answer is one number: skill + 82 x correct

- Histories cut into chunks of 200, padded, with a mask for the padding

- Split by student: 80% train, 20% test; validation is 10% of the train students



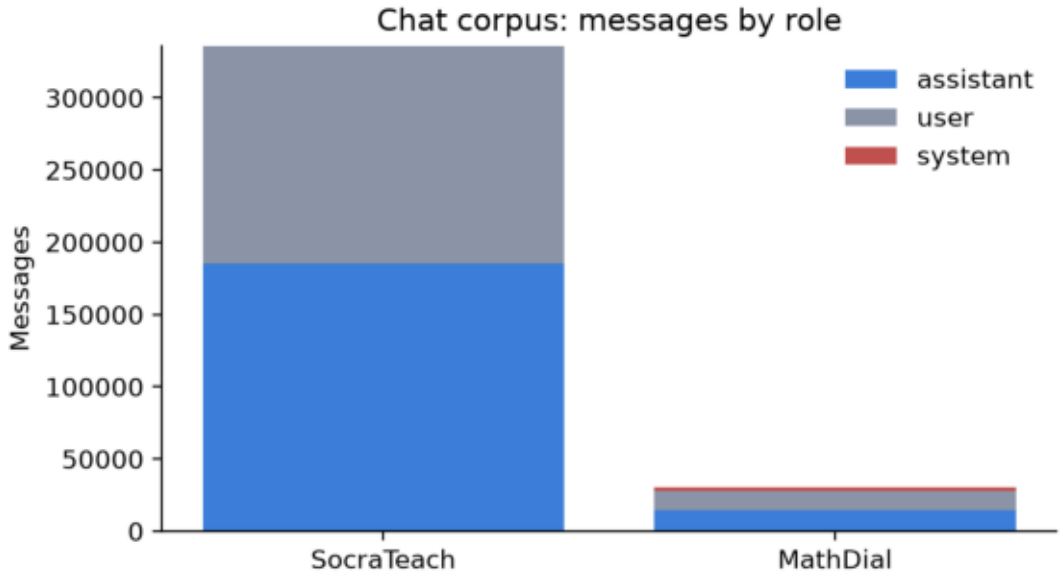
Chat corpus: Socratic model data

- 35,151 SocraTeach dialogues + 2,262 MathDial conversations

- Each conversation is a list of system, assistant and user messages, ready for fine-tuning

- MathDial's 599 test conversations are held out

- Validation: 215 MathDial conversations, chosen by problem



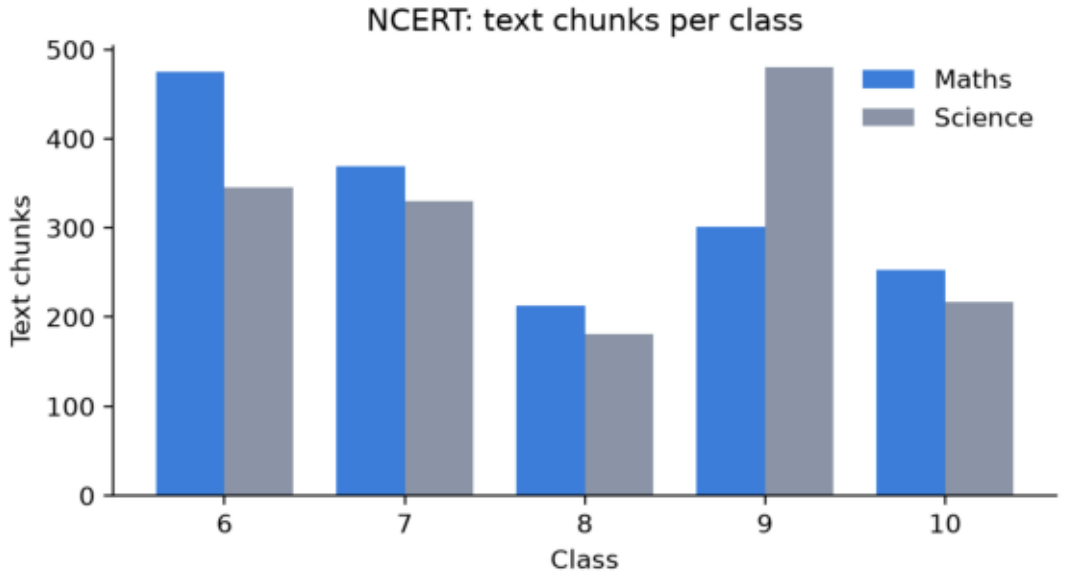
NCERT: study text

- Maths and Science, classes 6 to 10 (classes act as levels)

- 2,401 pages cleaned and cut into chunks of up to 250 words

- Each chunk keeps subject, class, chapter and page, so the tutor can point to the exact page

- Text stays local: the books are public but not ours to redistribute



Course packs: concepts and questions

- Hand-written chains of concepts that build across levels

- Example: percentages -> fractions -> multiplication and division

- Each concept links to its NCERT chapter, so the tutor can point to the page

- Practice questions have a hint and difficulty; hints never contain the answer

Pack	Concepts	Prerequisite links	Questions
NCERT Maths	10	13	52
NCERT Science	6	6	18

Our own test scenarios

- Backtracking: 82 cases (correct, careless, hint, gap one or two levels down)

- Misconceptions: 32 wrong answers with the error behind each

- Guardrails: 47 messages (on-topic, off-topic, answer request)

- Each backtracking case carries the learner's mastery, so cases with the same inputs always have the same label

- Never used for training or tuning; a second member reviews every label

Sample records

ASSISTments	`train,80959,chunk 0,step 1,skill 9,correct 1,x=91` (x = skill + 82 x correct)
Chat corpus	`{"source": "mathdial", "messages": [{"role": "system|assistant|user", "content": "..."}]}`
NCERT chunk	`{"subject": "maths", "class": 10, "chapter": 6, "page": 17, "text": "..."}`
Scenario	`{"type": "careless_mistake", "mastery": {"concept": 0.9}, "expected_first_action": "probe"}`
Misconception	`{"question": "What is 20% of 50?", "wrong_answer": "2.5", "misconception": "divides_by_the_percent"}`
Guardrail	`{"message": "What is the weather like today?", "label": "off_topic"}`

More samples and plots: Milestone2_EDA.ipynb

Processed dataset: folders and splits

```
hosted_dataset/
  assistments_dkt/    dkt_data.npz, dkt_processed_steps.csv, skill_map.csv
  socratic_chat/      socratic_tutor_combined.jsonl
  splits/             dkt_validation.json, chat_validation.json
  ncert_metadata/     chapters.csv (text not shared)
  course_pack_specs/  specs.py
  test_scenarios/     backtracking, misconceptions, guardrails

ASSISTments: train 2,453 / validation 273 / test 682 students (by student)
Chat corpus: train 37,198 / validation 215 (by problem) / MathDial test 599 (held out)
```

Hosted dataset (Google Drive): https://drive.google.com/drive/folders/15vIRFghBsqfHjvHrIztgvVSUv2vP4tws

Checks and limits

- ASSISTments checks: row counts add up, encoding holds on every step, no student in two splits

- Our checks pass: NCERT chunks, pack graphs, no hint contains its answer, consistent scenario labels

- ASSISTments is US data with no session information

- SocraTeach is non-commercial (CC BY-NC); chat data is maths only

- NCERT text: some digits lost in older books; study passages need a manual pass

Status against the Milestone 1 plan

Plan	Status	Note
NCERT as primary course	Done	text, chunks, 2 course packs
IITM BS pack	Not done	pipeline is course-agnostic; add later
Guardrail methods compared	Not done	eval set built (47); methods in M3
Controller / misconception baselines	Not started	moves to M3
Class 3-5 content (level 1)	Not done	only classes 6-10 processed

Next steps

- Second member reviews concepts, prerequisites and scenarios

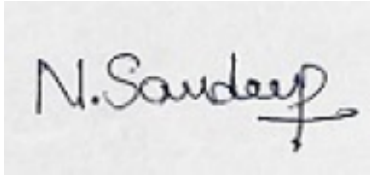
- Train DKT on ASSISTments; published baseline AUC 0.754 (pyKT benchmark, Liu et al. 2022)

- Add an answer-leak flag, then fine-tune the Socratic model on the chat corpus

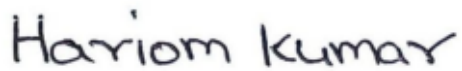
- Add the TA's test samples to our scenarios

Team sign-off

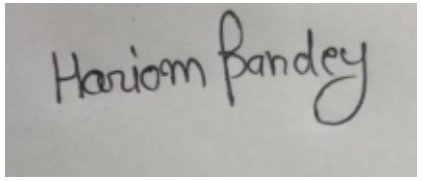
Sandeep Nalla (21f1006125)



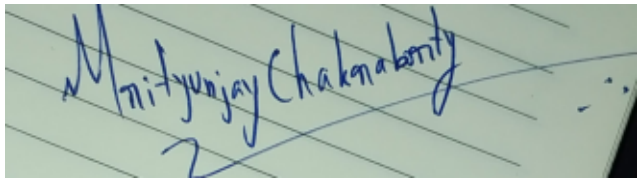
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Signed 29 October 2026